

CENTRAL KENTUCKY, KY, USA

WMO: 720455

Lat: 37.633N

Lon: 84.333W

Elev: 1001

StdP: 14.17

Time Zone: -5.00 (NAE)

Period: 11-19

WBAN: 00145

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.6	13.8	-1.9	5.1	12.7	2.9	6.6	15.5	24.6	42.2	22.3	38.7	3.9	330	0.416

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	91.3	73.5	89.8	73.1	87.6	72.7	76.6	86.4	75.4	84.6	74.2	83.3	6.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	127.8	81.7	72.4	124.6	81.0	71.9	122.4	80.5	40.6	86.3	39.3	85.0	38.5	83.1	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.0	17.5	15.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			1.2	78.7	9.0	1.3	-5.3	79.6	-10.6	80.3	-15.6	81.1	-22.2	82.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.9	34.4	39.4	46.1	57.5	66.5	73.2	76.3	74.0	69.0	57.9	46.3	41.2
	DBStd	17.02	12.74	12.43	11.49	9.44	8.33	5.26	4.89	4.35	7.38	9.15	9.89	10.27
	HDD50	1616	502	335	214	39	2	0	0	0	0	30	181	313
	HDD65	4209	949	718	589	255	89	6	1	1	43	255	564	739
	CDD50	4137	19	37	94	264	514	695	814	744	569	277	71	39
	CDD65	1255	0	1	4	30	136	252	350	280	162	36	4	0
	CDH74	10529	0	1	24	242	1022	2145	3091	2269	1439	284	12	0
	CDH80	3563	0	0	0	23	250	726	1222	750	536	55	1	0
Wind	WSAvg	5.8	7.4	7.1	7.1	7.3	5.7	5.0	3.9	3.6	3.8	5.5	6.0	6.8
Precipitation	PrecAvg	47.0	3.0	3.0	4.3	4.0	4.9	4.6	4.6	4.1	3.7	3.0	3.7	3.9
	PrecMax	63.5	8.0	8.5	10.6	10.8	9.9	9.6	8.9	9.4	10.0	7.2	8.9	12.9
	PrecMin	32.7	0.6	0.4	1.0	0.9	0.9	0.8	1.1	1.1	0.6	0.0	0.7	0.6
	PrecStd	8.2	1.7	1.8	2.4	2.1	2.1	2.1	1.8	2.0	2.2	1.7	1.8	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.7	72.3	78.6	82.2	88.4	93.2	98.5	92.7	95.1	85.7	75.0	68.1
		MCWB	58.9	61.2	62.5	64.8	71.7	70.8	73.1	72.4	72.0	71.3	62.5	60.8
	2%	DB	61.5	67.7	72.9	80.5	84.4	89.8	91.4	89.8	90.3	81.1	69.7	64.3
		MCWB	54.8	59.4	59.0	63.6	69.9	72.1	75.5	71.6	71.5	67.2	59.7	57.8
	5%	DB	58.5	63.0	68.4	77.0	82.3	87.5	89.7	87.5	86.2	77.2	64.4	61.2
		MCWB	53.0	55.1	57.1	61.8	68.4	71.9	75.3	72.7	71.1	64.5	55.9	55.6
	10%	DB	54.1	58.7	64.0	73.0	80.8	84.2	86.3	84.1	82.0	72.8	62.5	57.4
		MCWB	48.9	51.9	53.9	60.1	67.5	71.0	73.7	71.4	70.1	63.1	53.6	52.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.1	62.7	64.4	68.3	73.9	76.5	79.1	76.4	75.6	72.9	64.1	62.5
		MCDB	64.2	69.4	76.0	78.5	84.6	86.4	88.7	85.5	85.5	82.1	71.0	66.9
	2%	WB	57.2	60.1	61.6	66.3	71.8	74.8	77.5	75.3	74.0	70.3	62.0	59.3
		MCDB	61.0	65.5	70.0	77.2	82.3	84.9	87.9	84.5	84.8	76.4	67.4	63.6
	5%	WB	53.7	57.3	59.2	64.2	70.5	73.4	76.3	74.2	72.6	67.7	58.7	56.5
		MCDB	57.8	61.8	66.5	73.3	80.2	83.5	85.8	82.9	83.2	73.9	63.5	60.7
	10%	WB	49.7	53.4	56.6	62.2	68.7	72.2	75.0	73.0	71.2	64.3	54.9	53.7
		MCDB	54.0	56.9	62.9	70.7	77.6	81.6	83.8	81.3	80.7	72.5	60.3	57.6
Mean Daily Temperature Range	5% DB	MDBR	16.3	17.1	18.6	20.5	18.6	19.5	18.4	19.2	20.1	19.6	18.8	14.9
		MCDBR	19.4	21.1	22.1	24.2	20.5	23.1	21.0	22.5	25.0	23.6	19.7	19.1
		MCWBR	15.0	15.0	13.6	12.7	9.5	9.4	7.5	8.9	9.2	11.2	12.4	14.8
	5% WB	MCDBR	18.7	20.0	18.7	19.9	18.5	18.6	17.8	17.4	20.4	19.6	15.4	17.1
		MCWBR	15.3	16.4	12.8	12.4	9.1	9.1	7.9	8.0	8.6	11.3	11.4	14.0
Clear-Sky Solar Irradiance	taub	0.307	0.318	0.348	0.391	0.441	0.450	0.477	0.476	0.406	0.355	0.319	0.299	
	taud	2.478	2.431	2.400	2.280	2.177	2.189	2.130	2.123	2.293	2.431	2.492	2.510	
	Ebn at Noon	279	290	290	282	268	265	257	254	267	272	272	275	
	Edh at Noon	26	30	35	42	47	46	49	48	38	30	25	23	
All-Sky Solar Radiation	RadAvg	620	826	1184	1574	1761	1959	1835	1733	1437	1088	763	542	
	RadStd	69	120	102	117	164	103	96	94	154	134	82	53	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+216	+133

Nomenclature: See separate page